

QUIZ 1

5 questions and 1 min for each one

Which of the following best captures the definition of bioinformatics used in this course?

- A. A set of wet-lab procedures designed to generate millions of sequencing reads per sample.
- B. The disciplined use of computational, statistical, and biological reasoning to convert high-dimensional data into reproducible evidence.
- C. The exclusive use of software tools to store and organize large biological databases.
- D. The automated process of allowing artificial intelligence to interpret biological results without human oversight.

In the RStudio interface, which pane is primarily used to check which objects (like data frames or variables) are currently loaded in the workspace?

A. Files / Plots / Help

B. Environment

C. Console

D. Source

According to the course materials, what is the primary molecular unit measured in the 'Metabolomics' omics layer?

- A. Small molecules and metabolic flux
- B. Proteins and signaling output
- C. RNA expression and pathway activation
- D. DNA sequences and genetic variants

In the CARE+V framework for effective prompting, what does the 'V' stand for, and why is it included?

- A. Visualization; to ensure the AI generates a boxplot or heatmap.
- B. Variation; to account for the technical noise inherent in biological samples.
- C. Validation; to perform wet-lab experiments such as qPCR or Western blots.
- D. Verification; to ask the model to list assumptions and diagnostic checks.

A student wants to compare the expression of a gene between two groups. Which R operation is correctly used to store the result in an object named 'result'?

A. `result <- mean(group1) - mean(group2)`

B. `mean(group1) - mean(group2) -> result`

C. `result =! mean(group1) - mean(group2)`

D. `result == mean(group1) - mean(group2)`